

Locikline™ Film-Coated Tablets
Levofloxacin

QUALITATIVE AND QUANTITATIVE COMPOSITION

Levofloxacin, 250 mg, film-coated tablets

Each film-coated tablet contains levofloxacin hemihydrate equivalent to levofloxacin 250mg

Levofloxacin, 500 mg, film-coated tablets

Each film-coated tablet contains levofloxacin hemihydrate equivalent to levofloxacin 500mg

Excipients

Tablets

Microcrystalline cellulose (Avicel PH 101), corn starch, colloidal silicon dioxide (Aerosil 200), crospovidone, hypromellose 15 cPs (Methocel E15 LV premium), microcrystalline cellulose (Avicel PH 102), magnesium stearate, purified water, Opadry white OY58900

PHARMACEUTICAL FORM

Tablets

Levofloxacin tablets 250mg:

White coloured modified capsule shaped, biconvex, film-coated tablets debossed with 'RDY' on one side and '279' on other side.

Levofloxacin tablets 500mg:

White coloured modified capsule shaped, biconvex, film-coated tablets debossed with 'RDY' on one side and '280' on other side

CLINICAL INFORMATION

Indications

Tablets

For the treatment of mild to moderate infections caused by levofloxacin-susceptible microorganisms in adults (>18 years of age):

- ❖ Community-acquired pneumonia
- ❖ Complicated urinary tract infections (including pyelonephritis)
- ❖ Acute bacterial sinusitis.
- ❖ Acute bacterial exacerbations of chronic bronchitis

National and/or local guidelines on the appropriate use of fluoroquinolones should be considered before prescribing levofloxacin.

Dosage and Administration

Usual Dose: 250 or 500 mg administered orally over 60 min every 24 hrs, as indicated by infection and described in the following dosing chart. These recommendations apply to patients with normal renal function (ie, creatinine clearance >80 mL/min).

For patients with altered renal function, see Table 2. Oral doses should be administered at least 2 hrs before or 2 hrs after antacids containing magnesium, aluminum, as well as sucralate, metal cations eg, iron and multivitamin preparations with zinc or didanosine chewable/buffered tablets or the pediatric powder for oral solution.

Patients with Normal Renal Function: See Table 1.

Table 1				
Infection*	Unit Dose (mg)	Frequency	Duration** (days)	Daily Dose (mg)

Acute bacterial exacerbation of chronic bronchitis	500	every 24 hrs	7	500
Community-acquired pneumonia	500	every 24 hrs	7-14	500
Acute maxillary sinusitis	500	every 24 hrs	10-14	500
Uncomplicated skin and skin structure infections	500	every 24 hrs	7-10	500
Complicated UTI	250	every 24 hrs	10	250
Acute pyelonephritis	250	every 24 hrs	10	250
* Due to the designated pathogens (see Indications).				
**Sequential therapy (IV to oral) may be instituted at the discretion of the physician.				

Patients with Impaired Renal Function: See Table 2.

Table 2		
Renal Status	Initial Dose	Subsequent Dose
Acute bacterial exacerbation of chronic bronchitis/community-acquired pneumonia/acute maxillary sinusitis/uncomplicated skin and skin structure infections.		
CL _{CR} 50-80 mL/min	No dosage adjustment required.	
CL _{CR} 20-49 mL/min	500 mg	250 mg every 24 hrs
CL _{CR} 10-19 mL/min	500 mg	250 mg every 48 hrs
Hemodialysis	500 mg	250 mg every 48 hrs
CAPD	500 mg	250 mg every 48 hrs
Complicated UTI/acute pyelonephritis.		
CL _{CR} ≥ 20 mL/min	No dosage adjustment required.	
CL _{CR} 10-19 mL/min		
	250 mg	250mg every 48 hrs
Uncomplicated UTI	No dosage adjustment required	
CL _{CR} = Creatinine clearance. CAPD = Chronic ambulatory peritoneal dialysis.		

When only the serum creatinine is known, the following formula may be used to estimate creatinine clearance: See equation.

$$\text{Creatinine Clearance (mL/min)} = \frac{\text{Weight (kg)} \times (140 - \text{age})}{72 \times \text{serum creatinine (mg/dL)}}$$

Females: 0.85 x value calculated using formula for males.

The serum creatinine should represent a steady state of renal function.

Contraindications

Levofloxacin is contraindicated in:

- ❖ Hypersensitivity to levofloxacin or other quinolones or any of the excipients
- ❖ Patients with epilepsy
- ❖ Patients with a history of tendon disorders related to the administration of fluoroquinolones
- ❖ Children or growing adolescents
- ❖ Pregnancy and lactation

Warnings and Precautions

The efficacy and safety of levofloxacin in children, adolescents (<18 years), pregnancy and nursing women have not been established.

Levofloxacin may not be the optimal therapy for the most severe cases of pneumococcal pneumonia.

Nosocomial infections caused by *Pseudomonas aeruginosa* may require combination therapy.

Tendinitis and tendon rupture

Tendinitis occurs rarely. It most frequently involves the Achilles tendon and may lead to its rupturing. The risk of tendinitis and tendon rupture is increased in elderly patients and in patients taking corticosteroids. If these patients are prescribed levofloxacin, they should be closely monitored. Patients should consult their physician if they experience symptoms of tendinitis. If tendinitis is suspected, treatment with levofloxacin must be stopped immediately and specific measures for the affected tendon must be taken (e.g. immobilization).

Tendon rupture can occur within 48 hours of the start of treatment and be bilateral, muscular weakness which may be particularly significant in patients with myasthenia gravis.

***Clostridium difficile* infections**

If severe, persistent and/or bloody diarrhoea occurs during or after treatment with levofloxacin tablets, this could be symptomatic of *Clostridium difficile* disease, the most severe form of which is pseudomembranous colitis which, if suspected, requires that administration of levofloxacin be stopped immediately and patients be treated with supportive measures with or without specific therapy (e.g. oral vancomycin). In this clinical situation products inhibiting the peristalsis are contraindicated.

Patients prone to seizures

Levofloxacin is contraindicated in patients with a history of epilepsy (*see section Contraindications*) and, as with other quinolones, should be used with extreme care in patients prone to seizures, such as subjects with pre-existing lesions of the central nervous system, concomitant treatment with fenbufen and similar NSAIDs or with drugs like theophylline that lower the cerebral seizure threshold, *see section Interactions*. In case of convulsive seizures, treatment with levofloxacin should be stopped.

Patients with a deficiency of glucose-6-phosphate dehydrogenase

Patients with latent or confirmed defects in glucose-6-phosphate dehydrogenase activity may be prone to haemolytic reactions when they are treated with quinolone type antibacterial agents, therefore levofloxacin should be used with care.

Patients with impaired kidney function

Since levofloxacin is mainly excreted by the kidneys, levofloxacin doses should be adjusted for patients with impaired kidney function, *see section Dosage and Administration*.

Hypersensitivity reactions

Levofloxacin can cause serious, potentially fatal hypersensitivity reactions (e.g. angioedema to anaphylactic shock), occasionally after the initial dose, *see section Adverse Reactions*. Patients should immediately stop treatment and contact their physician or an emergency physician, who will begin appropriate emergency treatments.

Hypoglycaemia

Like with all quinolones, hypoglycaemia has been reported, ordinarily in diabetic patients who are receiving combination treatment with oral glucose-lowering drugs (e.g. glibenclamide) or with insulin. Careful monitoring of blood glucose levels is recommended in these diabetic patients, *see section Adverse Reactions*.

Prevention of photosensitivity

Even though photosensitivity is very rare, it is recommended that patients undergoing levofloxacin therapy not expose themselves unnecessarily to strong sunlight or artificial UV rays (e.g. sun lamp, solarium), in order to prevent photosensitivity.

Psychotic reactions

In patients taking quinolones, including levofloxacin, psychotic reactions have been reported. These, in very rare cases, have progressed to suicidal thoughts and self-harming behaviour - occasionally after only a single dose of levofloxacin, see section Adverse Reactions. If these reactions occur in the patient, levofloxacin should be discontinued and appropriate measures should be taken. Levofloxacin should be used with care in patients with a history of psychosis or psychiatric disorders.

QT interval prolongation

Fluoroquinolones, including levofloxacin, should be administered with care in patients with risk factors known for prolonging the QT interval such as:

- ❖ congenital long QT syndrome,
- ❖ simultaneous use of drugs that are known to prolong the QT interval, *see section Interactions*
- ❖ uncorrected electrolyte imbalance (e.g. hypokalemia, hypomagnesemia), the elderly,

❖ cardiac diseases (e.g. heart failure, myocardial infarction, bradycardia), see sections *Warning and Precautions, Interactions, Adverse Reactions & Overdosage*.

Peripheral neuropathy

Sensory or sensorimotor peripheral neuropathy has been reported in patients taking fluoroquinolones, including levofloxacin, which can have a rapid onset. If the patient has symptoms of neuropathy, levofloxacin treatment should be stopped in order to prevent the irreversible condition from developing.

Hepatobiliary disorders

After treatment with levofloxacin, cases of hepatic necrosis with potentially life threatening liver failure have been reported mostly in patients with severe underlying diseases, such as sepsis, see section *Adverse Reactions*. Patients should be told to discontinue treatment and contact their physician if signs and symptoms of liver disease appear such as anorexia, jaundice, dark urine, itching or tender abdomen.

Interactions

Theophylline, fenbufen or similar non-steroidal anti-inflammatory drugs

A clinical study demonstrated that there were no pharmacokinetic interactions between levofloxacin and theophylline. A marked reduction of the convulsion threshold is however possible when quinolones are administered concomitantly with theophylline, NSAID or other drugs that lower this threshold.

In the presence of fenbufen, levofloxacin concentrations were 13% higher than those observed when administering the drug by itself.

Probenecid and cimetidine

Particular care should be given when levofloxacin is administered concomitantly with drugs that modify tubular renal secretion such as probenecid and cimetidine, especially in patients with kidney impairments.

Ciclosporin

Ciclosporin's half-life increased by 33% when it was administered concomitantly with levofloxacin.

Vitamin K antagonists

Increases in the coagulation test values (PT/INR) and/or bleeding, which may be severe, have been reported in patients being treated with levofloxacin along with a vitamin K antagonist (e.g. warfarin). The coagulation tests should therefore be monitored in patients being treated with vitamin K antagonists.

Drugs known to prolong the QT interval

As with other fluoroquinolones, levofloxacin should be used with care in patients taking drugs known to prolong the QT interval (e.g. Class IA and III antiarrhythmics, tricyclic antidepressants, macrolides), see section *Warning and Precautions*.

Antidiabetic drugs

Like with all quinolones, hypoglycaemia has been reported, ordinarily in diabetic patients who are receiving combination treatment with oral glucose-lowering drugs (e.g. glibenclamide) or with insulin, see section *Warning and Precautions*.

Corticosteroids

The risk of tendinitis and tendon rupture is increased in patients taking concomitantly levofloxacin and corticosteroids, see section Warning and Precautions.

Opiates

The determination of opiates in the urine may give false-positive results in patients being treated with levofloxacin. A more specific method may be required in order to confirm the positive result.

Other types of interactions

Tablets

Iron salts, antacids containing magnesium or aluminium

The absorption of levofloxacin is significantly reduced when levofloxacin tablets are administered concomitantly with iron salts or antacids containing magnesium or aluminium. Preparations containing divalent or trivalent cations such as iron salts or antacids containing magnesium or aluminium should not be taken 2 hours before or after the administration of levofloxacin tablets. No interaction with calcium carbonate was found.

Sucralfate

When administered concomitantly with sucralfate, the bioavailability of levofloxacin tablets is significantly reduced. In the case of concomitant therapy, it is recommended that the sucralfate be administered 2 hours after the administration of levofloxacin tablets.

Food

Food has little effect on the absorption of levofloxacin.

Pregnancy and Lactation

Fertility

There are no relevant data available.

Pregnancy

Levofloxacin is contraindicated during pregnancy due to the lack of human data and because of the experimental risk of damage by fluoroquinolones to the weight-bearing cartilage of the growing organism, *see section Contraindications*. Reproductive studies on animals did not show any specific problems.

Lactation

Levofloxacin is contraindicated in women who are breast-feeding due to the lack of human data and because of the experimental risk of damage by fluoroquinolones to the weight-bearing cartilage of the growing organism, *see section Contraindications*.

Ability to perform tasks that require judgement, motor or cognitive skills

Some undesirable effects (e.g. dizziness/vertigo, drowsiness, visual disorders) may reduce the patient's ability to concentrate and react, creating a subsequent risk in situations where these abilities are very important (while driving a car or operating machinery).

Adverse Reactions

Infections and infestations

Uncommon: Fungal infection (and proliferation of other resistant microorganisms)

Blood and lymphatic system disorders

Uncommon: leukopenia, eosinophilia

Rare: thrombocytopenia, neutropenia

Very rare: agranulocytosis

Not known: pancytopenia, haemolytic anaemia

Immune system disorders

Very rare: anaphylactic shock, *see section Warning and Precautions*

Not known: hypersensitivity, *see Skin and subcutaneous tissue disorders and section Warning and Precautions*

Anaphylactic and anaphylactoid reactions may even occasionally occur after the first dose.

Metabolism and nutritional disorders

Uncommon: anorexia

Very rare: hypoglycaemia, especially in diabetic patients, *see section Warning and Precautions*

Psychiatric disorders

Uncommon: insomnia, nervousness

Rare: psychotic reactions, depression, confusion, restlessness, anxiety

Very rare: psychotic reactions with self-harming behaviour including suicidal thoughts or acts *see section Warning and Precautions*, hallucinations

Nervous system disorders

Uncommon: dizziness, headache, drowsiness

Rare: convulsions, tremors, paraesthesia

Very rare: sensory or sensorimotor peripheral neuropathy, dysgeusia including ageusia, parosmia including anosmia

Eye disorders

Very rare: visual disturbances

Ear and labyrinth disorders

Uncommon: vertigo

Very rare: reduced hearing

Not known: tinnitus

Cardiac disorders

Rare: tachycardia

Not known: QT interval prolonged, *see section Warning and Precautions* Overdosage.

Vascular disorders

Rare: hypotension

Respiratory, thoracic and mediastinal disorders

Rare: bronchospasm, dyspnoea

Very rare: allergic pneumonitis

Gastrointestinal disorders

Common: diarrhoea, nausea

Uncommon: vomiting, abdominal pain, indigestion, flatulence, constipation

Rare: diarrhoea bloody which in very rare cases can be a sign of enterocolitis, including pseudomembranous colitis

Hepatobiliary disorders

Common: hepatic enzymes increased (ALT - AST, alkaline phosphatase, (GGT)

Uncommon: blood bilirubin increased

Very rare: hepatitis

Not known: jaundice and severe liver damage, including cases of acute liver failure, have been reported with levofloxacin, mainly in patients with severe pre-existing diseases, see section Warning and Precautions.

Skin and subcutaneous tissue disorders

Uncommon: rash, itching

Rare: urticaria

Very rare: angioneurotic oedema, photosensitivity reactions

Not known: toxic epidermal necrolysis, Stevens-Johnson syndrome, erythema multiforme, hyperhidrosis

Mucocutaneous reactions may even occasionally occur after the first dose, see section Warning and Precautions.

Musculoskeletal and connective tissue disorders

Rare: tendon disorders including tendinitis (e.g. Achilles tendon) see section Warning and Precautions, arthralgia, myalgia

Very rare: tendon rupture, see section Warning and Precautions

Not known: rhabdomyolysis

Renal and urinary disorders

Uncommon: increased blood creatinine

Very rare: acute renal insufficiency (e.g. due to interstitial nephritis)

General disorders and administration site conditions

Uncommon: asthenia

Very rare: pyrexia

Not known: pain (including backache, chest and limb pain)

Overdosage

Symptoms and signs

Based on toxicity studies on animals and clinical pharmacology studies conducted at higher than therapeutic doses, the most significant symptoms that can be expected after an acute overdose of levofloxacin include: confusion, dizziness, lowering of the state of consciousness, convulsions and prolongation of the QT interval. Nausea and mucosal erosions can occur after overdosage with levofloxacin tablets.

Treatment

Symptomatic treatment should be implemented in the case of an overdose. Electrocardiographic monitoring should be performed due to the possibility of QT interval prolongation. Haemodialysis, including continuous ambulatory peritoneal dialysis (CAPD), are not effective in eliminating levofloxacin. There is no specific antidote.

Clinical Pharmacology

Pharmacodynamics

Pharmacotherapeutic group

Quinolone antibacterials, fluoroquinolones

ATC Code

J01MA12

Mechanism of Action

Levofloxacin tablet is a synthetic broad spectrum antibacterial fluoroquinolone containing levofloxacin which is the S (-) enantiomer (levorotatory form) of the racemic drug substance ofloxacin for oral and intravenous administration.

Levofloxacin acts on the DNA-DNA gyrase complex by inhibiting DNA gyrase (bacterial topoisomerase II), an enzyme required for DNA replication, transcription, repair and recombination, and topoisomerase IV.

Mechanism of resistance

The main mechanism of resistance is caused by a gyr-A mutation. In vitro there is cross-resistance between levofloxacin and other fluoroquinolones. Generally there is no cross-resistance between levofloxacin and other classes of antibacterial agents for this particular mechanism of action

Pharmacodynamic effects

PK/PD relationship

The level of levofloxacin's antibacterial activity depends on the ratio between the maximum serum concentration (C_{max}) or the area under the curve (AUC) and the minimal inhibitory concentration (MIC).

Breakpoints

The MIC breakpoint values (mg/l) for levofloxacin recommended by EUCAST, which separates susceptible from intermediately susceptible organisms and intermediately susceptible from resistant organisms, are reported in the Table 4:

Table 4 EUCAST recommended clinical breakpoints for levofloxacin, expressed in MIC (20-06-2006)

Pathogen	Susceptible	Resistant
Enterobacteriaceae	≤1 mg/l	> 2 mg/l
Pseudomonas spp.	≤1 mg/l	> 2 mg/l
Acinetobacter spp.	≤1 mg/l	>2 mg/l
Staphylococcus spp.	≤1 mg/l	> 2 mg/l
S. pneumoniae ¹	≤2 mg/l	>2 mg/l
Streptococcus A, B, C, G	≤1 mg/l	>2 mg/l
H. influenzae M. catarrhalis ²	≤1 mg/l	> 2 mg/l
Non-species related breakpoints ³	≤1 mg/l	> 2 mg/l
1 The S/I breakpoint was increased from 1.0 to 2.0 to prevent the "wild type" MIC distribution from dividing. The breakpoints are related to high dose therapy. 2 Strains with MIC values greater than the S/I breakpoint are very rare or have not yet been reported. In these situations, the identification and antimicrobial susceptibility tests must be repeated on each isolate strain and if the result is confirmed the strain must be sent to a reference laboratory. 3 Non-species related breakpoint values have mainly been determined based on the pharmacokinetic/pharmacodynamic data and are independent of the MIC distributions of specific species. These values are only to be used for species where a specific breakpoint has not been found and are not for those species where susceptibility testing is not recommended or where there is insufficient evidence that the species in question are a good target (Enterococcus, Neisseria, Gram negative anaerobic bacteria)		

Reported in the table below are the CLSI (*Clinical and Laboratory Standard Institute*, formerly called NCCLS) recommended MIC breakpoints for levofloxacin, which separates susceptible from intermediately susceptible organisms and intermediately susceptible from resistant organisms, expressed in MIC values (g/ml) and disc diffusion testing (diameter of the inhibition zone in mm using a 5 g disc of levofloxacin).

Table 5 MIC and disc diffusion breakpoints for levofloxacin recommended by CLSI (M100-S17, 2007):

Pathogen	Susceptible	Resistant
Enterobacteriaceae	≤ 2 µg/ml ≥ 17 mm	≥ 8 µg/ml ≤13 mm
Non Enterobacteriaceae	≤ 2 µg/ml ≥ 17 mm	≥ 8 µg/ml ≤13 mm
Acinetobacter spp.	≤ 2 µg/ml ≥ 17 mm	≥ 8 µg/ml ≤13 mm
Stenotrophomonas maltophilia	≤ 2 µg/ml ≥ 17 mm	≥ 8 µg/ml ≤13 mm
Staphylococcus spp.	≤ 1 µg/ml ≥ 19 mm	≥ 4 µg/ml ≤15 mm
Enterococcus spp.	≤ 2 µg/ml ≥ 17 mm	≥ 8 µg/ml ≤13 mm
H. influenzae M. catarrhalis ¹	≤ 2 µg/ml ≥ 17 mm	
Streptococcus pneumoniae	≤ 2 µg/ml ≥ 17 mm	≥ 8 µg/ml ≤13 mm
beta-hemolytic Streptococcus	≤ 2 µg/ml ≥ 17 mm	≥ 8 µg/ml ≤13 m

1 The absence or rare occurrence of resistant strains prevents any result categories other than "susceptible" from being defined. For strains giving results suggesting a "non susceptible" category, the results of the microorganism identification and antimicrobial susceptibility tests must be confirmed by a reference laboratory that uses the CLSI standard dilution method.

Antibacterial spectrum

The prevalence of resistance can vary with time and place. Local information on resistance should be obtained, especially when treating severe infections. An expert should be consulted in situations when the local prevalence of resistance is such that the usefulness of the product in at least some types of infections is in doubt.

Commonly susceptible species Gram-positive Aerobic Bacteria <i>Staphylococcus aureus</i> * methicillin-susceptible <i>Staphylococcus saprophyticus</i> Group C and G streptococci, <i>Streptococcus agalactiae</i> <i>Streptococcus pneumoniae</i> * <i>Streptococcus pyogenes</i> * Gram-negative Aerobic Bacteria <i>Burkholderia cepacia</i> **
<i>Eikenella corrodens</i>, <i>Haemophilus influenzae</i> * <i>Haemophilus para-influenzae</i> * <i>Klebsiella oxytoca</i> <i>Klebsiella pneumoniae</i> *

***Moraxella catarrhalis* ***

Pasteurella multocida

Proteus vulgaris

Providencia rettgeri

Anaerobic Bacteria

Peptostreptococcus

Other changes

Chlamydophila pneumoniae *

Chlamydophila psittaci

Chlamydia trachomatis

Legionella pneumophila *

Mycoplasma pneumoniae *

Mycoplasma hominis

Ureaplasma urealyticum

Species for which acquired resistance may be a problem

Gram-positive Aerobic Bacteria

Enterococcus faecalis *

Staphylococcus aureus methicillin-resistant

Staphylococcus spp coagulase negative

Gram-negative Aerobic Bacteria

Acinetobacter baumannii *

Citrobacter freundii *

Enterobacter aerogenes

Enterobacter agglomerans

Enterobacter cloacae *

Escherichia coli *

Morganella morganii *

*Proteus mirabilis**

Providencia stuartii,

Pseudomonas aeruginosa *

Serratia marcescens *

Anaerobic Bacteria

***Bacteroides fragilis*,**

Bacteroides ovatus**

Bacteroides thetaiotamicron* *

Bacteroides vulgatus **

Clostridium difficile **

* The clinical efficacy has been shown in the isolated susceptible strains in the approved clinical indications.

** natural intermediate susceptibility

Other information

Nosocomial infections caused by *P. aeruginosa* may require combination therapy.

Pharmacokinetics

Absorption

When administered orally, levofloxacin is rapidly and almost completely absorbed with a peak plasma concentration being obtained in 1 hour. The absolute bioavailability is approximately 100%.

Food has little effect on the absorption of levofloxacin.

Distribution

Approximately 30 - 40 % of levofloxacin is bound to serum protein. Levofloxacin showed negligible accumulation with 500 mg daily multiple dosing. There is however modest but predictable accumulation of levofloxacin after 500 mg doses administered twice a day. Steady-state is achieved within approximately 3 days.

Penetration into tissues and body fluids:

Penetration into Bronchial Mucosa, Epithelial Lining Fluid

The maximum concentrations of levofloxacin in bronchial mucosa and epithelial lining fluid are 8.3µg/g and 10.8µg/ml respectively after the oral administration of 500 mg and were reached one hour after administration.

Penetration into lung tissue

The maximum concentration of levofloxacin in lung tissue after the oral administration of 500 mg was approximately 11.3µg/g. This concentration was reached 4-6 hours after administration and consistently exceeds that in plasma.

Penetration into blister fluid

The maximum concentration of levofloxacin (approximately 4 and 6.7 µg/ml) in blister fluid was reached 2-4 hours after administration and 3 days after treatment with 500 mg once or twice a day, respectively.

Penetration into cerebrospinal fluid

Levofloxacin has poor penetration into cerebrospinal fluid.

Penetration into prostatic tissue

After oral administration of 500mg of levofloxacin once a day for three days, the mean concentrations in prostatic tissue were 8.7 µg/g, 8.2 µg/g and 2.0 µg/g after 2, 6 and 24 hours respectively; the mean ratio of prostate/plasma concentrations was 1.84.

Concentration in urine

8 -12 hours after a single oral dose of 150, 300 or 500 mg of levofloxacin, the mean urine concentrations were 44 mg/l, 91 mg/l and 200 mg/l, respectively.

Metabolism

Levofloxacin is metabolised to a small extent into the desmethyl-levofloxacin and levofloxacin N-oxide metabolites. These metabolites account for < 5% of the amount excreted into the urine. Levofloxacin is stable from a stereochemical point of view and does not undergo chiral inversion.

Elimination

Levofloxacin is eliminated slowly from the plasma after oral or intravenous administration ($t_{1/2}$: 6-8 hours).

Excretion is primarily done through the kidneys (> 85% of the administered dose).

The plasma concentration profile of intravenous levofloxacin is similar and comparable in extent of exposure to that observed for equal doses of tablets of levofloxacin; therefore the oral and intravenous routes of administration can be considered interchangeable.

Linearity

Levofloxacin follows linear pharmacokinetics over dose range from 50 to 600 mg.

Special patient populations

Elderly

There are no significant kinetic differences in levofloxacin between young and elderly patients, except those associated with differences in creatinine clearance, *see section Warning and Precautions (QT interval prolongation)*.

Renal impairment

The pharmacokinetics of levofloxacin are influenced by impaired kidney function. Renal elimination and clearance are reduced with decreasing kidney function, and elimination half-lives increase as reported in the following table:

Cl _{CR} (ml/min)	< 20	20-40	50-80
Cl _R (ml/min)	13	26	57
t _{1/2} (h)	35	27	9

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Other patient characteristics

Separate testing on male and female subjects showed small to marginal differences in the pharmacokinetics of levofloxacin. There is no evidence relating to the clinical relevance of these differences.

PHARMACEUTICAL INFORMATION

Shelf-Life

The expiry date is indicated on the packaging.

Storage

Store below 30°C.

Nature and Contents of Container

PVC/Alu

Incompatibilities

There are no relevant data available.

Use and Handling

There are no special requirements for use or handling of this product.

Not all presentations are available in every country.

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LOCIKLINE is a trademark of the GlaxoSmithKline group of companies.

Manufactured by:

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